Xu Han

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Photonics master's student and LMU astrophysics graduate focused on astronomical instrumentation, combining experimental apparatus development, optical modelling and quantitative research.

Education

M.Sc. Photonics | Munich University of Applied Sciences *Mar 2026 - Present*
Selected coursework: Physical Optics and Modelling (**Zemax optical design**); Micro and Fiber Optics; Optical Measurement Technology; Semiconductor Optics and Components.

M.Sc. Astrophysics | LMU Munich *Apr 2022 - Aug 2024*
Grade: 1.55 (German scale). **Selected coursework:** Essentials of Advanced Astrophysics; Cosmology and Structure Formation; Statistical Methods; Instrumental Laboratory Courses.

B.Sc. Physics | University of Duisburg-Essen *Oct 2018 - Mar 2022*
Grade: 1.8 (German scale).

B.Eng. Electrical Engineering and Automation *Sep 2013 - Jul 2017*
Hubei University of Technology

Selected Research and Engineering Projects

Shack-Hartmann AO Simulation *Mar 2026 - Present*

Independent optical modelling project (ongoing development)

- Developing a modular Python SCAO framework connecting atmospheric turbulence, Shack-Hartmann detector images, calibrated wavefront reconstruction, deformable-mirror correction and delayed feedback.
- Investigated detector noise, reconstruction regularisation and gain/latency trade-offs using residual wavefront error, Strehl ratio, FWHM and encircled energy.
- Checked physical conventions and numerical results against analytic references and independent HCIPy optics; documented reproducible experiments in executed notebooks.

Experimental Collisional Charging Apparatus *Mar 2021 - Aug 2021*

Bachelor's thesis | University of Duisburg-Essen
Supervisors: Prof. Gerhard Wurm and Dr. Felix Jungmann.

- Designed the mechanical apparatus in CAD, fabricated components by 3D printing and assembled the setup; integrated and debugged electronic charge-measurement modules.
- Conducted measurements and analysed grain-size dependence of collisional charging, testing a dust-collision growth hypothesis relevant to early planet formation.

Weak Lensing Void Finder *Mar 2023 - Aug 2024*

Master's thesis | LMU Munich
Supervisors: Prof. Joseph Mohr and Dr. Chris Davies.

- Developed and parameter-tested a watershed finder for cosmic voids in 2D full-sky weak-lensing convergence maps.
- Applied the analysis to DES Y3 weak-lensing maps; characterised void abundance, effective radii, ellipticities and convergence profiles.

Laboratory and Instrumentation Training

- **Photonics laboratory:** APD measurements, laser spectra, nonlinear optics, laser interferometry, fiber Bragg gratings and fiber coupling.
- **Astronomical instrumentation laboratory:** CCD imaging and data reduction, optical interferometry, deformable-mirror AO tuning and radio measurements of Galactic rotation.
- **Observatory experience:** Participated in observing and image processing at Wendelstein Observatory using the 43 cm telescope.

Additional Independent Projects

Astrophotonics

- Modelled how AO wavefront error, injection jitter and photonic-lantern mode count affect fiber coupling, spectral SNR and radial-velocity precision.

AO-Fed IFU Spectrograph Simulation

- Modelled the optical-to-detector chain and datacube reconstruction; compared recovered flux, velocity and SNR with synthetic truth and examined calibration sensitivity.

Optoelectronic Signal Chain Simulations

- Studied photodetector/APD noise, laser thermal control, LiDAR detection, camera sensors and spectrometer calibration through parameter scans and uncertainty analysis.

Work Experience

Python Developer | BaryonBib

Dec 2024 - Feb 2025

MAX!mize start-up team, MPA, Garching

- Contributed to a centralised bibliography platform in Python; implemented and tested LLM-based functionality.

Talks, Workshops and Schools

MAVIS 2026 workshop *Sep 2026*

Tonale Winter School on Cosmology *Dec 2024*

ORIGINS Lensing Day *Nov 2024*

Watershed on the curved sky

Cambridge-LMU Cosmology Meeting *Dec 2023*

Advanced Astrophysical Seminar | LMU Munich *Nov 2022*

40-minute course presentation and literature study on large ground-based astronomical telescopes, telescope designs and adaptive optics.

Technical Skills

Optical modelling: Wavefronts, Fourier PSFs, detector noise, calibration and inverse problems; HCIPy and TIPTOP; Zemax through optical-design coursework.

Scientific computing: Python (NumPy, SciPy, Matplotlib, pandas, scikit-learn); MATLAB; Mathematica; statistical analysis and MCMC.

Experimental methods: CAD, 3D printing, mechanical assembly, electronic measurement integration, optical alignment and interferometry.

Research tools: Linux, Git, Jupyter and LaTeX.

Languages and Interests

Languages: Chinese (native); English (fluent); German (fluent); French (beginner).

Interests: Astronomical photography and alpine mountaineering.